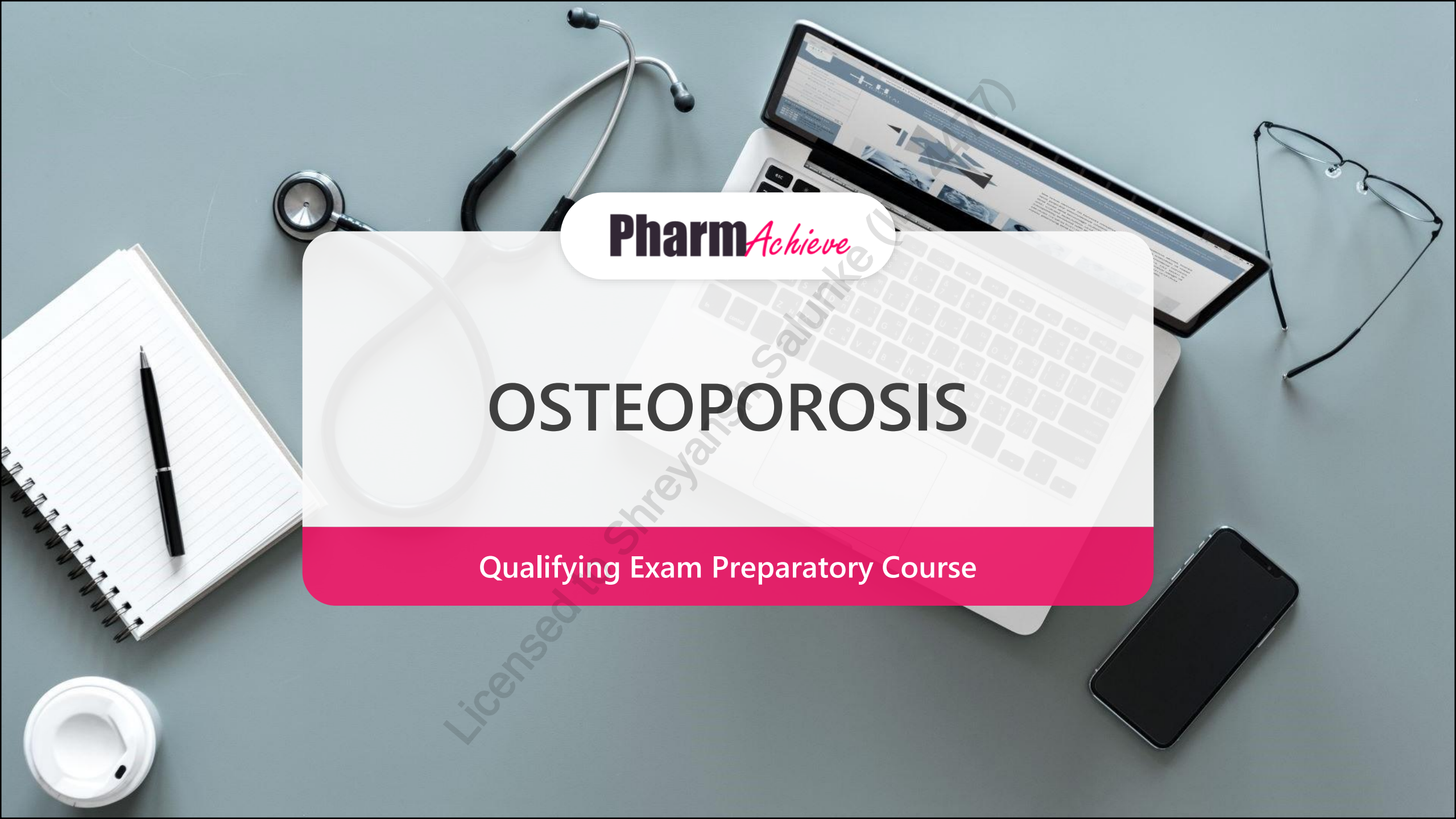


A top-down view of a grey desk with various medical and study items: a stethoscope, a laptop displaying a medical website, a tablet, a pair of glasses, a smartphone, a spiral notebook with a pen, and a coffee cup. A large white pill-shaped graphic is centered over the laptop.

Pharm*Achieve*

OSTEOPOROSIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

BMD	Bone Mineral Density	PO	Per Os (Oral)
CBC	Complete Blood Count	PPI	Proton Pump Inhibitor
CrCl	Creatinine Clearance	RANK	Receptor Activator of Nuclear Factor Kappa
COPD	Chronic Obstructive Pulmonary Disease	RANKL	RANK Ligand
DVT	Deep Vein Thrombosis	SC	Subcutaneous Injection
GI	Gastrointestinal	SERM	Selective Estrogen Receptor Modulator
IU	International Units	PMO	Post Menopausal Osteoporosis
LMWH	Low Molecular Weight Heparin	PTH	Parathyroid Hormone
LTC	Long-term Care	TSH	Thyroid Stimulating Hormone
mAb	Monoclonal Antibody	UFH	Unfractionated Heparin
OD	Once Daily	VTE	Venous Thromboembolism
ONJ	Osteonecrosis Of The Jaw	YR	Year
PE	Pulmonary Embolism		

DEFINITIONS

Osteoporosis	• Increased bone resorption and ↓ formation - leads to ↓ bone mass, thinning of bone, weaker bones • Patients are at risk of bone fractures
Osteoclasts	• Cells that degrade old and weak bone • Stimulated by calcitriol, parathyroid hormone and interleukins
Osteoblasts	• Cells that form connective tissue matrix that mineralizes and becomes bone
Osteocyte	• Main cell type in bones, longest-living • Controls activity of osteoblasts and osteoclasts
Bone matrix	• Organic and inorganic materials (mainly collagen for flexibility)

DEFINITIONS

DEXA	• Dual-Energy X-ray Absorptiometry • Most accurate bone density test
T-SCORE	• Comparison between patient's bone density to the bone density of the average young adult (World Health Organization reference) • Number of standard deviations below average
Z-SCORE	• Comparison of a patient's bone density to a reference database of same gender and age • Useful in investigating secondary osteoporosis • Used for children, youth, pre-menopausal women, and men < 50 years
Fragility Fracture	• Fracture that occurs due to falling from standing height OR • Fracture that occurs when force is applied to bone that would NOT normally cause a fracture
Major osteoporotic fracture	• Fracture of the hip, vertebra, humerus, distal forearm

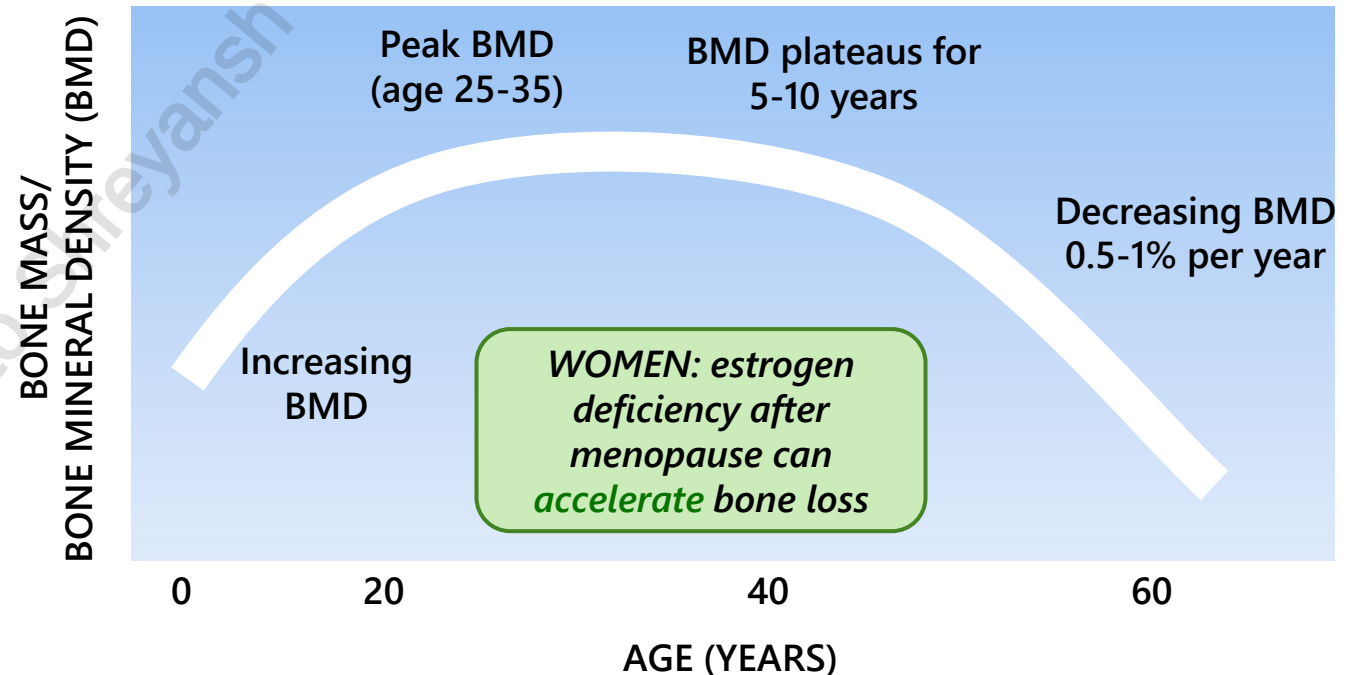
CLINICAL PRESENTATION

- Primary osteoporosis: age-related reduction in sex hormones (incidence greater in women than men), calcium, and vitamin D deficiencies
- Secondary osteoporosis: disease-related (Cushing's, rheumatoid arthritis)/drug (glucocorticoids)

Clinical Presentation:

- Many patients find out they have osteoporosis after a fracture
- Fractures can occur after bending, lifting, falling or independently of any activity
- Signs: shortened stature, kyphosis (rounding of upper back)/lordosis (inward curve of lumbar spine), pain, immobility
- Note: 2/3 vertebral fractures are asymptomatic

CHANGES IN BMD RELATIVE TO AGE



DIAGNOSIS

A patient can be diagnosed with osteoporosis if they are ≥ 50 years old

AND

Have had a low-trauma hip, vertebral, humerus or pelvic fracture after age 40-years-old

OR

If they have an absolute fracture risk of $\geq 20\%$ over the next 10 years using CAROC or FRAX



ASSESSMENT

Risk Factors for Fracture

- History of fragility fracture after age 40
- Glucocorticoids (> 3 months in the last year, prednisone dose > 5 mg daily)
- Falls, ≥ 2 in the last year
- Parent fractured hip
- Body mass index < 20 kg/m²
- Current smoking
- Alcohol ≥ 3 drinks/day
- *Causes of secondary osteoporosis

Causes of Secondary Osteoporosis

- Drugs such as lamivudine, ritonavir, phenytoin, valproate, carbamazepine, phenobarbital, furosemide, bumetanide, UFH, LMWH, long-term PPI use, excessive thyroid hormone supplementation
- Rheumatoid arthritis
- Aromatase inhibitors
- Systemic erythematous lupus
- Inflammatory bowel disease
- Chronic liver/kidney disease
- COPD
- Parkinson's disease
- Osteomalacia
- Malabsorptive syndromes

MEASURING 10-YEAR FRACTURE RISK

FRAX AND CAROC TOOLS

- Stratification into 10-year fracture risk of major osteoporotic fracture

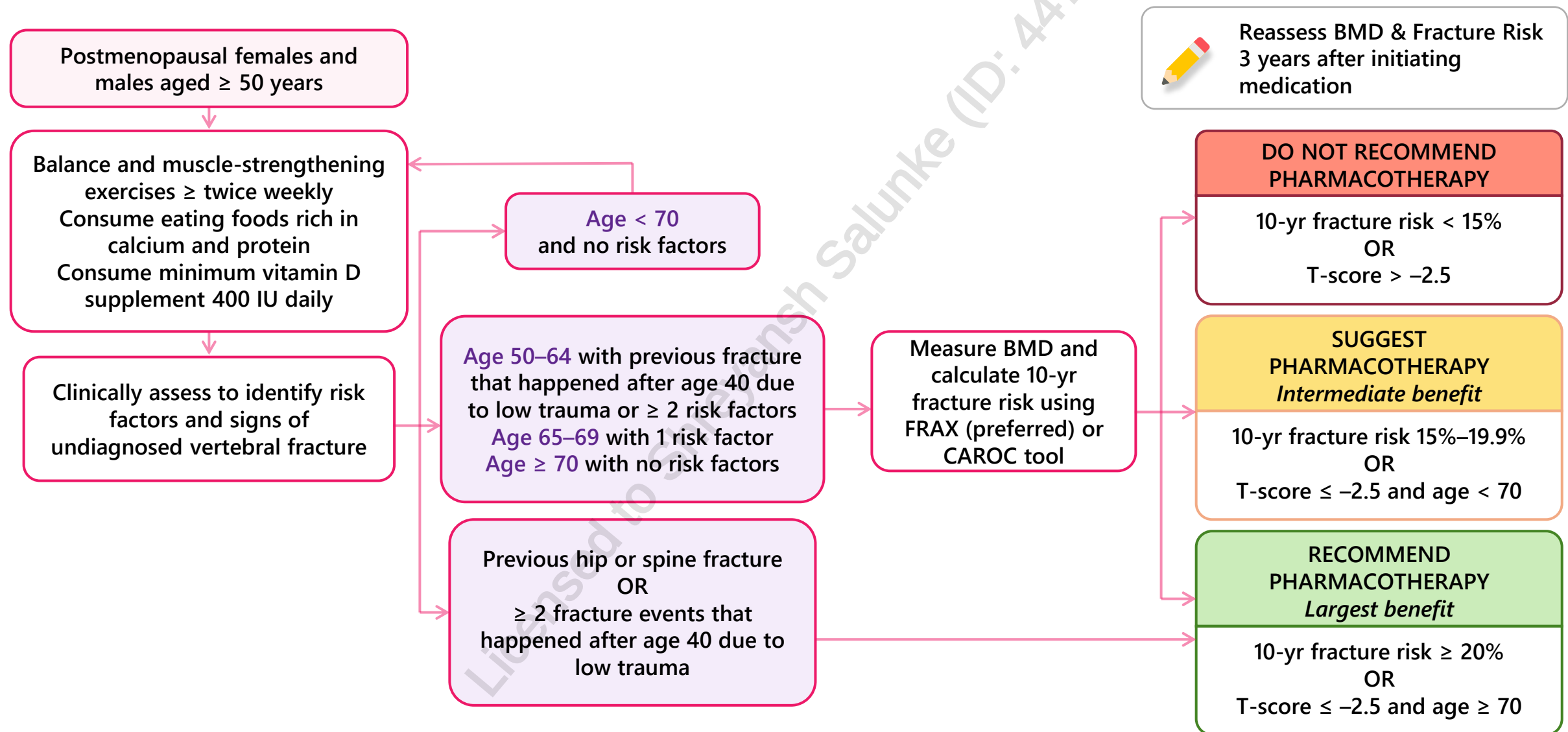
FRAX (Fracture Risk Assessment Tool) *preferred in Canada

- Age
- Sex
- BMI
- Prior fracture
- Parental hip fracture
- Prolonged steroid use
- Secondary causes of OP
- Smoking status
- Alcohol intake
- T-score for femoral neck (optional)

CAROC (Canadian Association of Radiologists and Osteoporosis Canada)

- Age
- Sex
- T-score for femoral neck

ASSESSMENT ALGORITHM



ADDITIONAL INVESTIGATIONS

- History of falls, visual acuity, disability
- Physical examination: kyphosis, muscle weakness, poor balance, height loss^{*}
 - ^{*} Height loss of ≥ 6 cm of peak adult height or ≥ 2 cm within past year could indicate a spine fracture. Everyone aged 50+ should have an annual height measurement.
- Lab testing: TSH, alkaline phosphatase, phosphate, CBC, creatinine, calcium (corrected for albumin), serum protein electrophoresis (for those with vertebral fractures), 25-Hydroxyvitamin D (should be measured approximately 3-4 months after sufficient supplementation – does not need to be repeated if optimal level achieved at that point)



Bone loss related to aging is normal!
The goal of these tests is to detect accelerated bone loss.

GOALS OF THERAPY

- ✓ Reduce the risk of fractures, osteoporosis, disability and loss of independence
- ✓ Preserve or enhance bone mass
- ✓ Identify patient's baseline risk factors for decreased BMD

NON-PHARMACOLOGICAL MEASURES

Exercise

- Balance and functional training ≥ 2 times/week to reduce risk of falls
- Progressive resistance training ≥ 2 times/week
- Aim for ≥ 150 minutes/week of moderate/vigorous physical activity but prioritize the above forms

Nutrition

- Daily Calcium intake:
 - Females >50 years: 1200 mg/day
 - Males 51-70 years: 1000 mg/day
 - Males >70 years: 1200 mg/day
- Limit alcohol (any alcohol intake is harmful)
- Offer smoking cessation support
- Protein in males and females >50 years: 0.8g/kg body weight/day



Patients consuming a balanced diet that are not being treated with medications for osteoporosis experience little to no benefit or detrimental effect on fractures with supplementation with calcium, vitamin D and protein

NON-PHARMACOLOGICAL MEASURES

Nutrition

Vitamin D increases calcium absorption

Vitamin D₃ (cholecalciferol) > vitamin D₂ (ergocalciferol)

Daily Vitamin D intake allowance

Osteoporosis Guidelines

- Ages 51-70: 600 IU/day
- Ages >70: 800 IU/day



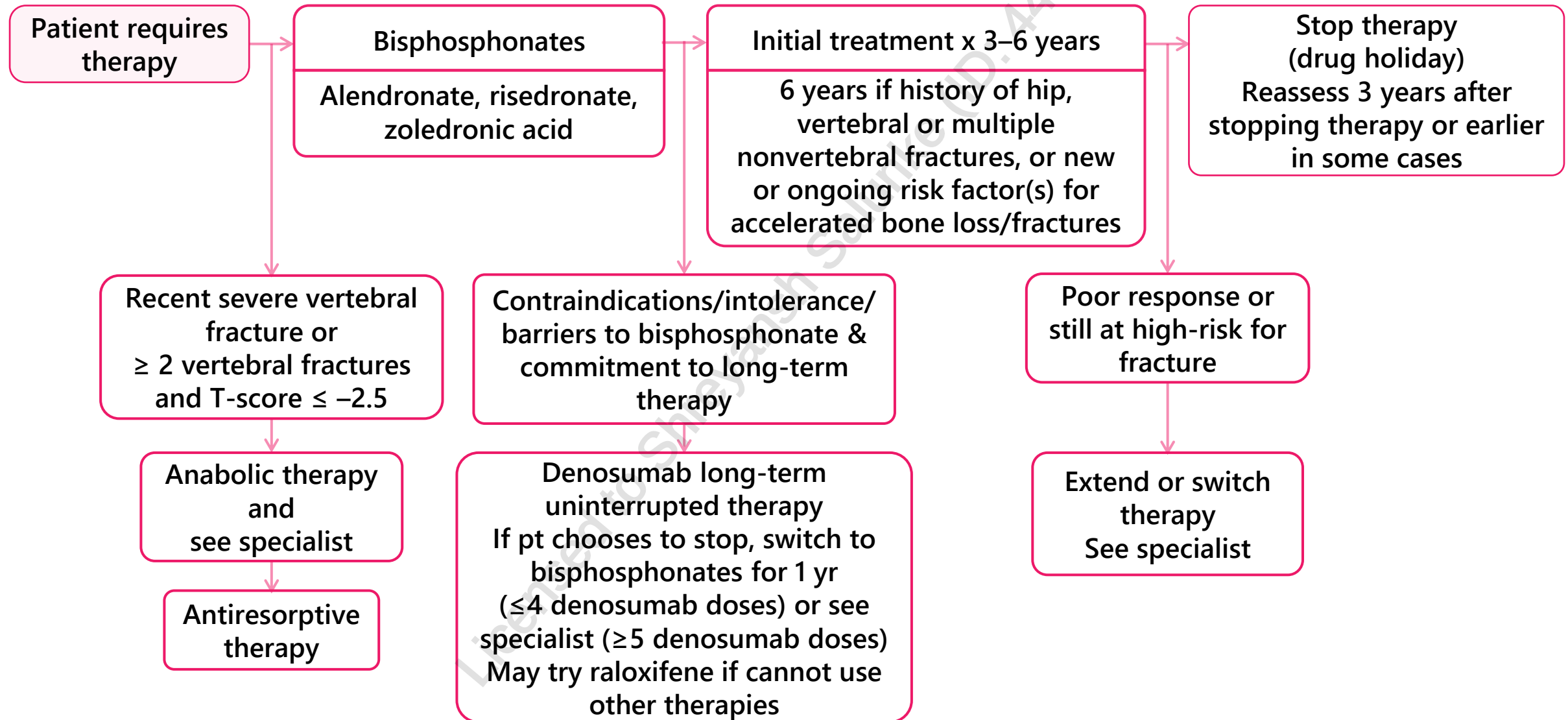
Adults >50 years should take a Vitamin D 400 IU supplement in addition to consuming Vitamin D rich foods as it is difficult to obtain enough vitamin D from diet alone. If at higher risk of deficiency, may need higher dose.

FALLS PREVENTION

FYI

- Hip protectors and falls alarms considered for those living in LTC facilities who are at a high risk for fracture
- Consider home safety assessment if patient has visual impairment or other high risks for falls
- Consider ongoing need for drugs that increase falls risk such as benzodiazepines
- Encourage exercise
- Install handlebars and rails near toilet and shower

TREATMENT ALGORITHM



PHARMACOLOGICAL THERAPY

Bisphosphonates	Denosumab	Romosozumab	Raloxifene	Menopausal Hormone therapy	Teriparatide
Alendronate: 70 mg PO once weekly or 10 mg PO daily Risedronate: 35 mg PO once weekly or 150 mg PO once monthly or 5 mg PO daily Zoledronic acid: 5 mg IV annually	60 mg SC every 6 months	Dose: 210 mg SC once monthly for 1 year	60 mg PO daily	Varies based on regimen	Dose: 20 mcg SC daily for 2 years
Antiresorptive	Antiresorptive	Anabolic	Antiresorptive	Antiresorptive	Anabolic
Prevents hip, vertebral and nonvertebral fractures <u>Mechanism:</u> Binds to bone mineral, gets retained in the skeleton and inhibits osteoclast activity and causes osteoclast apoptosis	Prevents hip, vertebral and nonvertebral fractures <u>Mechanism:</u> mAb; binds to RANKL, preventing it from binding to RANK, thus inhibiting osteoclast creation and survival	Prevents hip, vertebral and nonvertebral fractures more than bisphosphonates <u>Mechanism:</u> mAb; inhibits sclerostin, increases bone formation and decreases resorption	Prevents vertebral fractures <u>Mechanism:</u> Selective Estrogen Receptor Modulator (SERM); Estrogen agonist in bone to prevent bone loss. Estrogen antagonist in breast and uterus	Prevents hip, vertebral and nonvertebral fractures more than bisphosphonates <u>Mechanism:</u> Estrogen prevents bone loss	Prevents hip, vertebral and nonvertebral fractures more than bisphosphonates <u>Mechanism:</u> PTH analogue; stimulates osteoblasts, increases calcium absorption in GIT and renal tubular reabsorption of calcium

PHARMACOLOGICAL THERAPY

Bisphosphonates (alendronic acid PO, risedronate PO, zoledronic acid IV)	Denosumab SC	Romosozumab	Raloxifene	Menopausal Hormone therapy	Teriparatide
<u>Adverse effects</u> zoledronic acid: GI upset/flu-like symptoms, renal toxicity, hypocalcemia <u>Others:</u> esophageal ulceration <u>RARE:</u> osteonecrosis of the jaw, atypical femoral fractures when used > 6 years Contraindicated in CrCl < 30-35	<u>Adverse effects:</u> peripheral edema, hypocalcemia, musculoskeletal pain, dermatitis, risk of infections (e.g. cellulitis) <u>RARE:</u> increased atypical femur fractures and osteonecrosis of the jaw	<u>Adverse effects:</u> increased risk myocardial infarction, stroke, hypocalcemia, musculoskeletal pain <u>RARE:</u> Atypical femur fractures, osteonecrosis of the jaw	<u>Adverse effects:</u> increased risk of VTE, cardiovascular disease, stroke, vaginal bleeding, vasomotor symptoms, peripheral edema, flu-like symptoms, leg cramps/muscle spasms, weight gain, vulvovaginal mycotic infections	<u>Adverse effects:</u> increased risk of VTE, cardiovascular disease, stroke, breast cancer	<u>Adverse effects:</u> hypercalcemia, hypercalciuria, orthostatic hypotension, nausea, musculoskeletal pain Only use for 24 months due to risk of osteosarcoma
<u>Counselling:</u> Take oral bisphosphonates on an empty stomach 30 min before the first meal with water only and remain upright for 30 min. Other drugs should be avoided for > 30-60 mins. Minerals and dairy impair absorption if taken close together Delayed-release formulation available (taken with food)	<u>Counselling:</u> Caution warranted in severe renal impairment Rapid bone loss and risk of vertebral fractures if delayed dose or with discontinuation	<u>Counselling:</u> Inadequate vitamin D increases risk for hypocalcemia Caution warranted in severe renal impairment	<u>Counselling:</u> For postmenopausal women	<u>Counselling:</u> For postmenopausal women	<u>Counselling:</u> Caution warranted with active or previous kidney stone disease

SUMMARY OF TREATMENT RECOMMENDATIONS

- For people who meet criteria for initiation of pharmacotherapy → recommend bisphosphonates (oral preferred due to barriers to infusion with Zoledronic acid)
- For people meeting criteria for initiation of pharmacotherapy who have contraindications, substantial intolerance or barriers to bisphosphonates → recommend denosumab
- For postmenopausal females aged < 60 years or within 10 years of menopause initiating pharmacotherapy who prioritize alleviation of substantial menopausal symptoms → recommend menopausal hormone therapy as an alternative to bisphosphonates
 - The choice will also depend on individualized risks of menopausal hormone therapy, which consists of an estrogen dose equivalent of conjugated equine estrogens of 0.625 mg daily (plus progestogen in those with an intact uterus)
 - For postmenopausal females initiating pharmacotherapy who have contraindications or substantial intolerance to, or who choose not to take other suggested therapies → recommend raloxifene rather than no treatment
- For people meeting criteria for initiation of pharmacotherapy who have had a recent severe vertebral fracture, or ≥ 2 vertebral fractures, AND a T-score ≤ -2.5, → consult specialist for anabolic therapy (teriparatide or romosozumab)

MONITORING

- If patient **is being treated with drugs**, reassess BMD and fracture risk in 3 years
- If patient **is not being treated with drugs**:
 - If 10-yr fracture risk < 10% , reassess in 5–10 years
 - If 10-yr fracture risk 10%–15%, reassess in 5 years
 - If 10-yr fracture risk > 15%, reassess in 3 years
- Reassess sooner if patient develops secondary osteoporosis
- Ongoing monitoring is required for falls, fractures and side effects of medication

TAKEAWAY

- Prevention is key: aim to maximize peak bone density through calcium and vitamin D intake and exercise
- Main causes of osteoporosis are old age, reduced estrogen, and calcium and vitamin D deficiency
- **Calcium: Females >50 and males >70 years: 1200 mg/day. Males 51-70 years: 1000 mg/day**
- **All adults >50 years should have vitamin D 400 IU supplement/day**
- FRAX preferred over CAROC
- **In general, first-line therapy:** bisphosphonates for 3-6 years → if inadequate response or concern for fracture extend therapy or switch in case of side effects
- **First-line therapy if ≥ 2 vertebral fractures and T score ≤ -2.5 :** anabolic therapy
- Reassess BMD and FRAX in 3 years if having pharmacologic treatment

CASE 1

OP is a 63-year-old female who has had a hip fracture. She is quite confused and anxious, and she wants to do everything she can to ensure she will not have another fracture. Her past medical history is significant for rheumatoid arthritis and anxiety. She takes prednisone 12.5 mg PO daily with breakfast and lorazepam 0.5 mg PO TID PRN. She smokes 6-8 cigarettes a day and drinks 2-3 glasses of wine with dinner. She is not allergic to anything. She enjoys doing Tai Chi twice weekly and swims on the weekends. She works as an administrative assistant at a small accounting firm. She spends a lot of her time at her desk and has started wearing glasses to help with eye strain from the computer.

1. Which of the following is NOT considered a risk factor that should have prompted a measurement of OP's BMD?
 - a) Smoking status
 - b) Alcohol intake
 - c) Anxiety
 - d) Corticosteroid use



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2. How much Vitamin D should she be having per day from all sources?

- a) 400 IU/day
- b) 600 IU/day
- c) 800 IU/day
- d) 1000 IU/day



CASE 1

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3. BMD testing shows that OP has a T-score of -2.5 and her 10-year fracture risk is 18%. Which of the following would be an appropriate first-line pharmacologic treatment option for her?
- a) Teriparatide
 - b) Romosozumab
 - c) Raloxifene
 - d) Alendronate



CASE 1

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4. Which of the following is correct when it comes to counselling patients starting on a bisphosphonate?
- a) Take on an empty stomach 30 minutes before the first meal
 - b) Take on a full stomach 30 minutes after first meal of the day
 - c) Take on a full stomach lying down after a meal
 - d) Take with a full glass of water and lie down within 30 minutes of taking



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CHANGE LOG

August 2025

- Minor formatting and grammatical changes made throughout

September 2025

- Slides 4-5: Removed 'FYI'
- Slide 7: Added diagnosis slide

November 2025

- Slide 8: Reworded first bullet point in risk factors
- Slide 11: Added height loss to physical examination and associated note
- Slide 21: Minor update to takeaway
- Slide 26: Updated references

April 2026

- Slide 2: Updated NAPRA competencies
- Slide 19: Minor rephrasing of last bullet point: ≥ 2 vertebral fractures instead of > 1 vertebral fracture. Both terms are technically the same, but to maintain consistency with the wording on algorithm slide 16